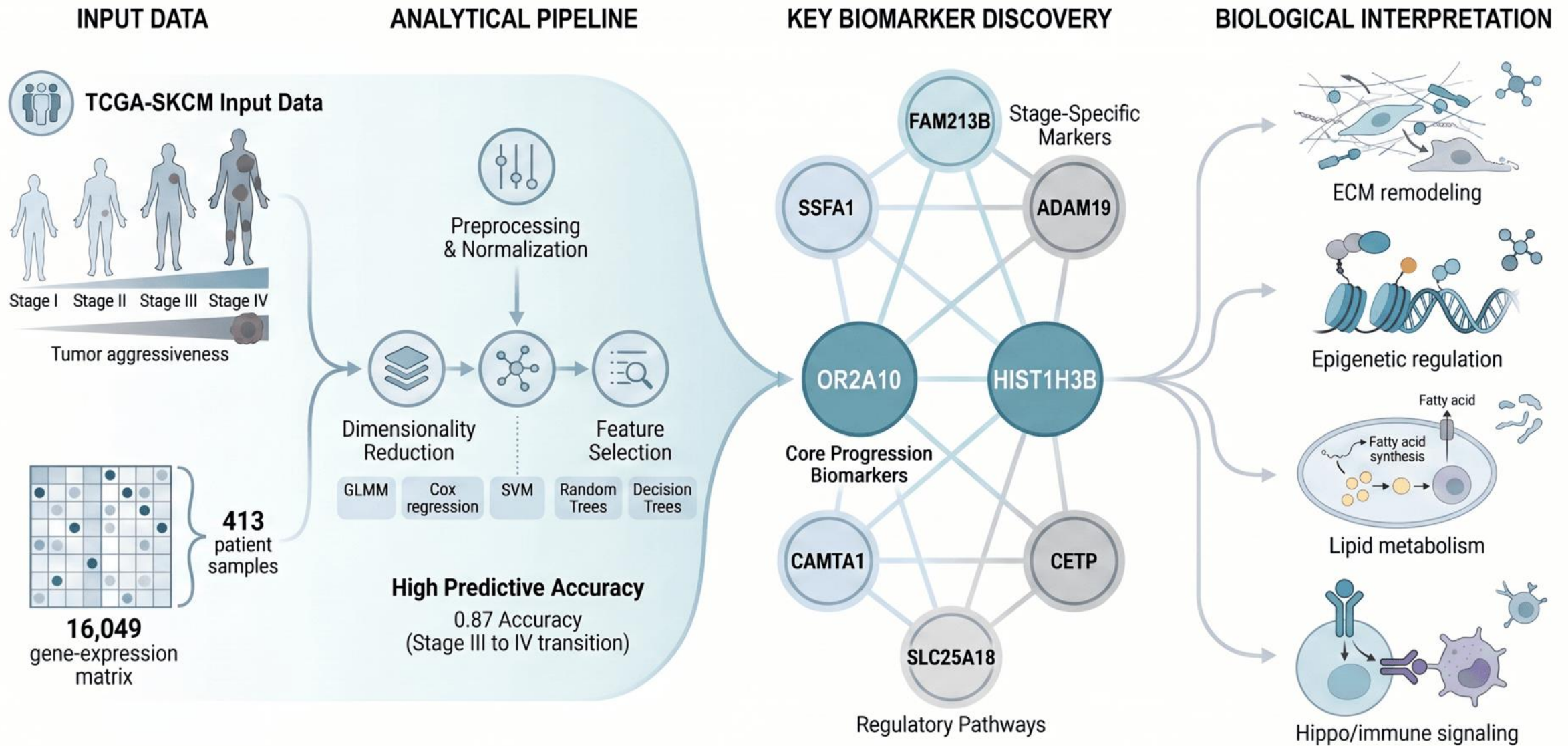


Supplementary

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Graphical abstract of stage-wise biomarker discovery in skin cancer progression



Graphical Abstract. Overview of the analytical workflow, key biomarkers, and biological pathways associated with skin cancer progression.

Proposed analytical workflow for identifying stage-associated biomarkers in skin cancer progression

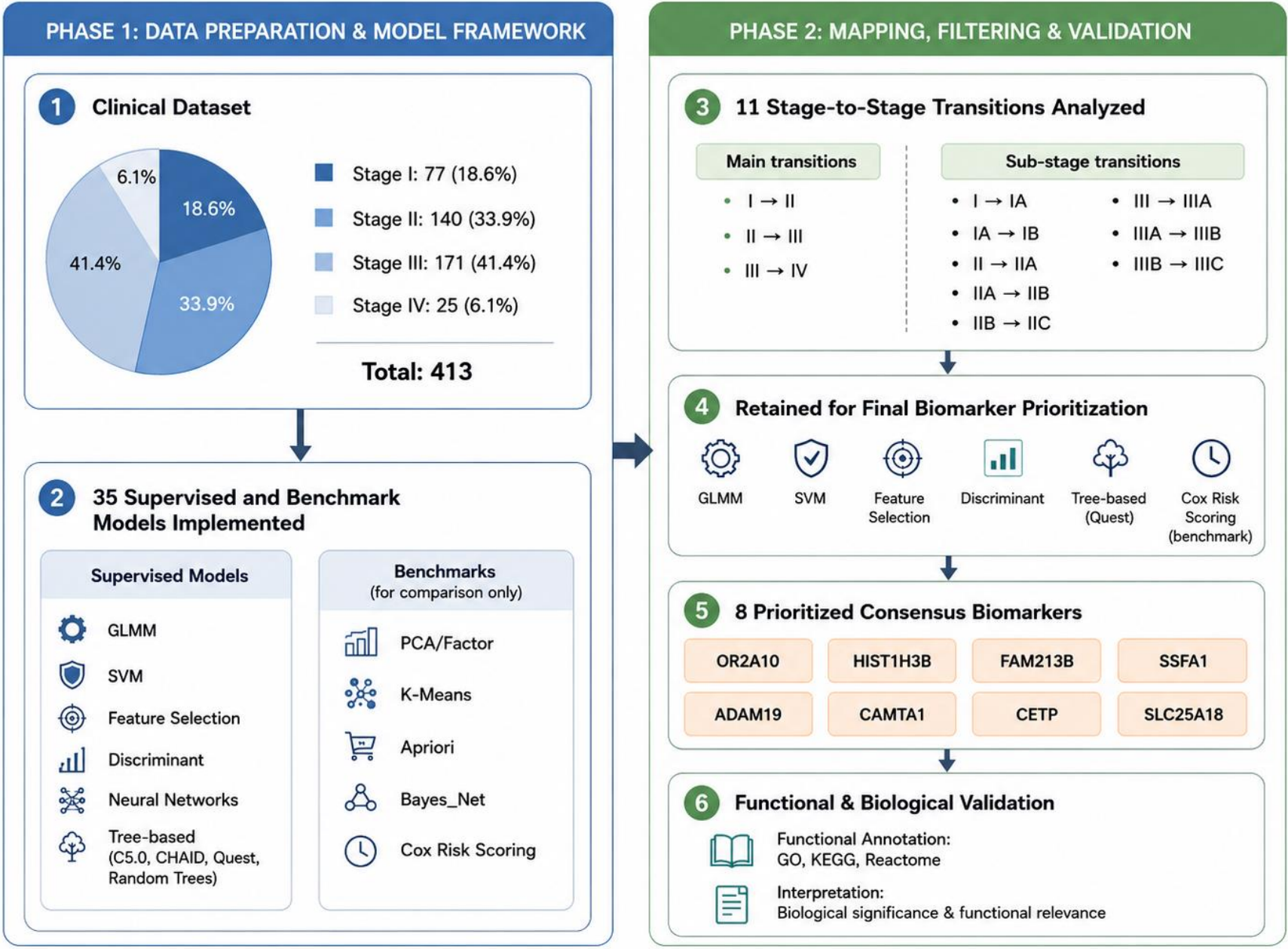


Figure 1. Proposed analytical workflow for identifying stage-associated biomarkers in skin cancer progression.

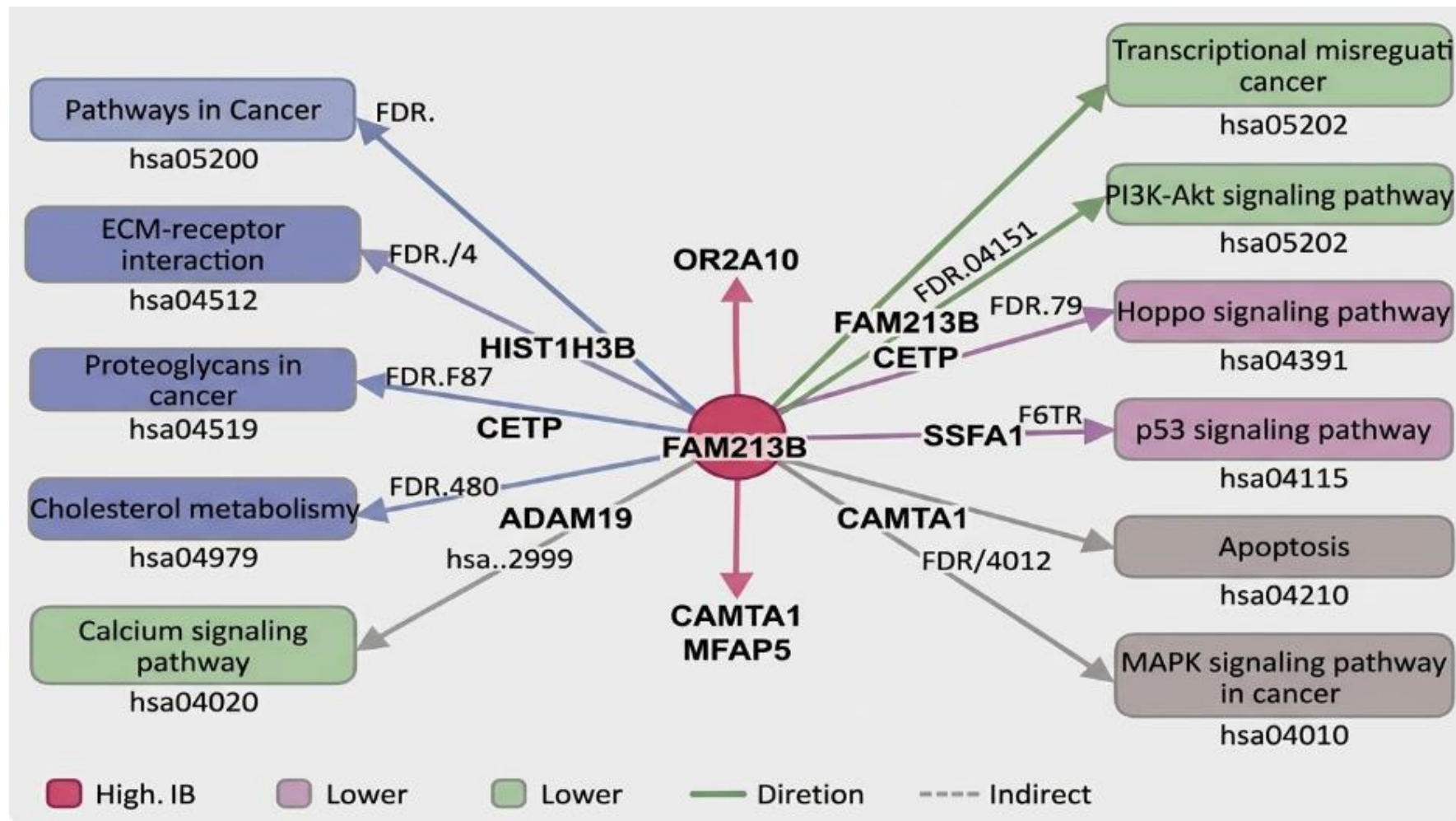


Figure 2. Integrated functional interaction network of the prioritized biomarkers based on functional enrichment evidence from multiple biological annotation databases. Representative gene–pathway associations are shown, while the complete enrichment results are provided in Supplementary Table S1.

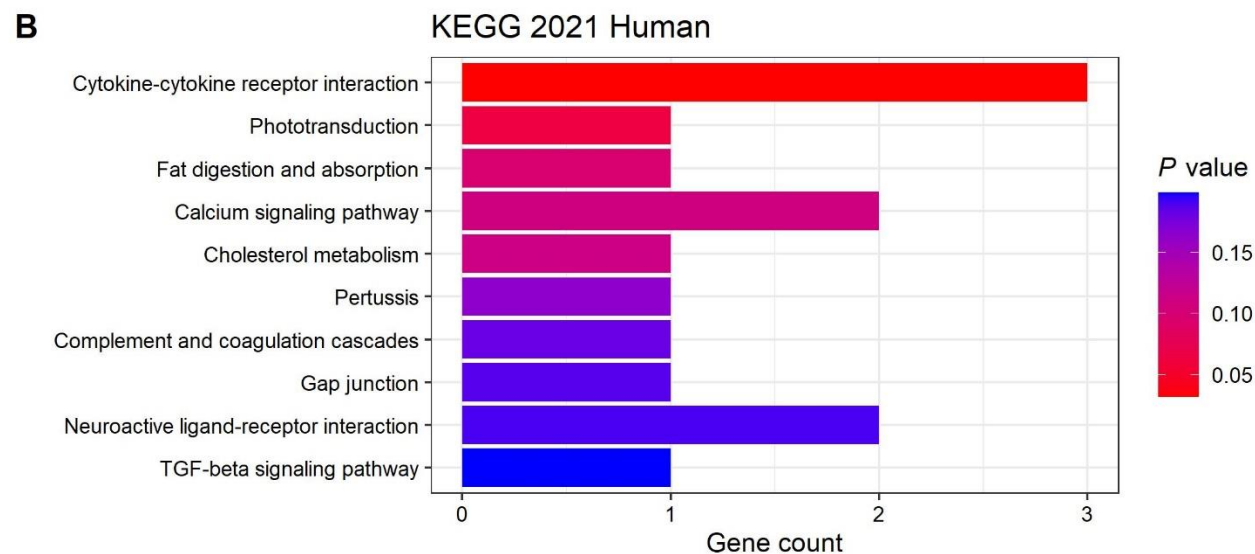
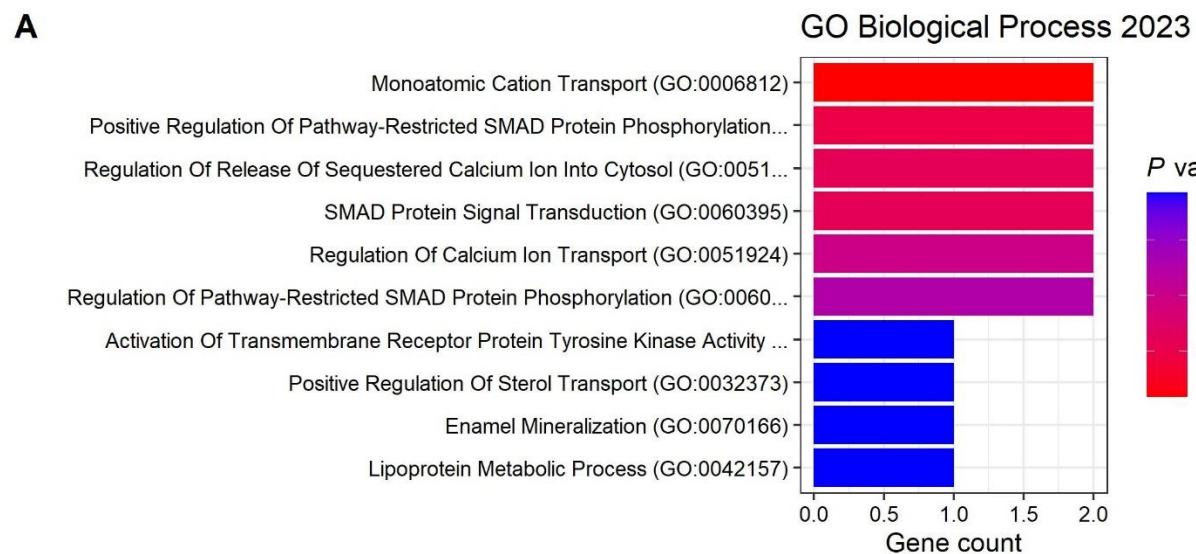


Figure 3. (A) Top enriched GO Biological Process terms (selected by adjusted p-value < 0.25 and combined score).

Figure 3. (B) Significantly enriched KEGG pathways associated with the prioritized genes.

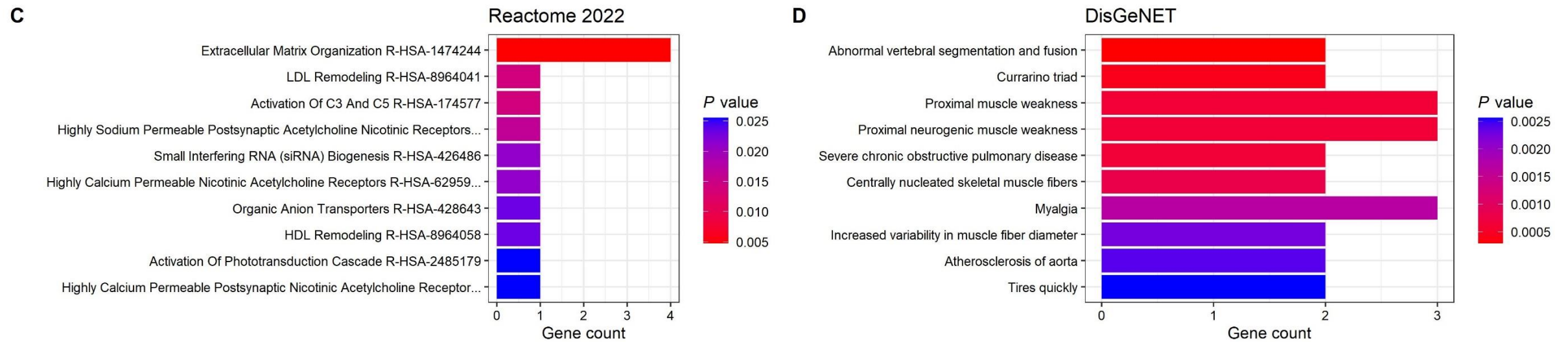


Figure 3. (C) Reactome enrichment results for key biomarkers.

Figure 3. (D) DisGeNET disease and phenotype associations for the prioritized genes.

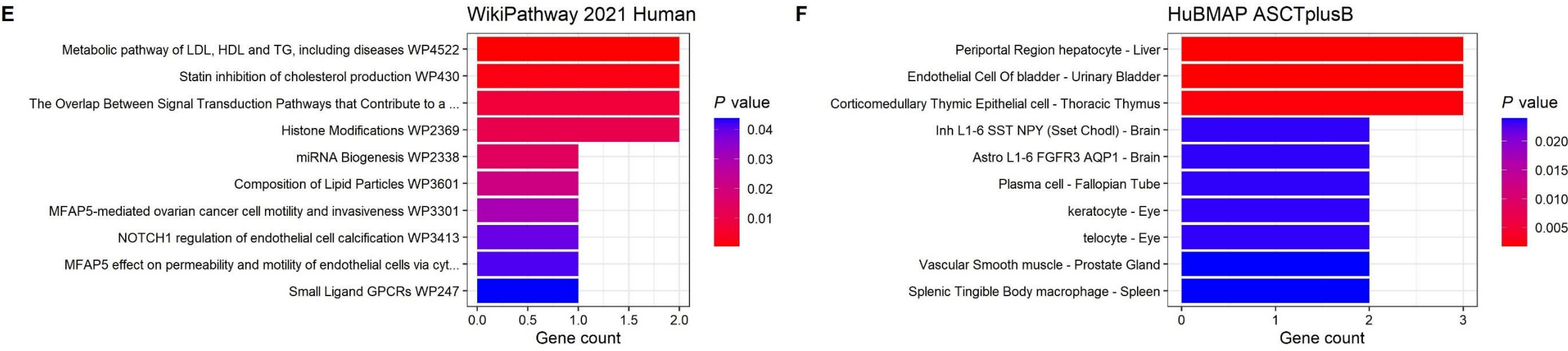


Figure 3. (E) WikiPathways enrichment statistics.

Figure 3. (F) HuBMAP cell-type distribution of stage-associated biomarkers.

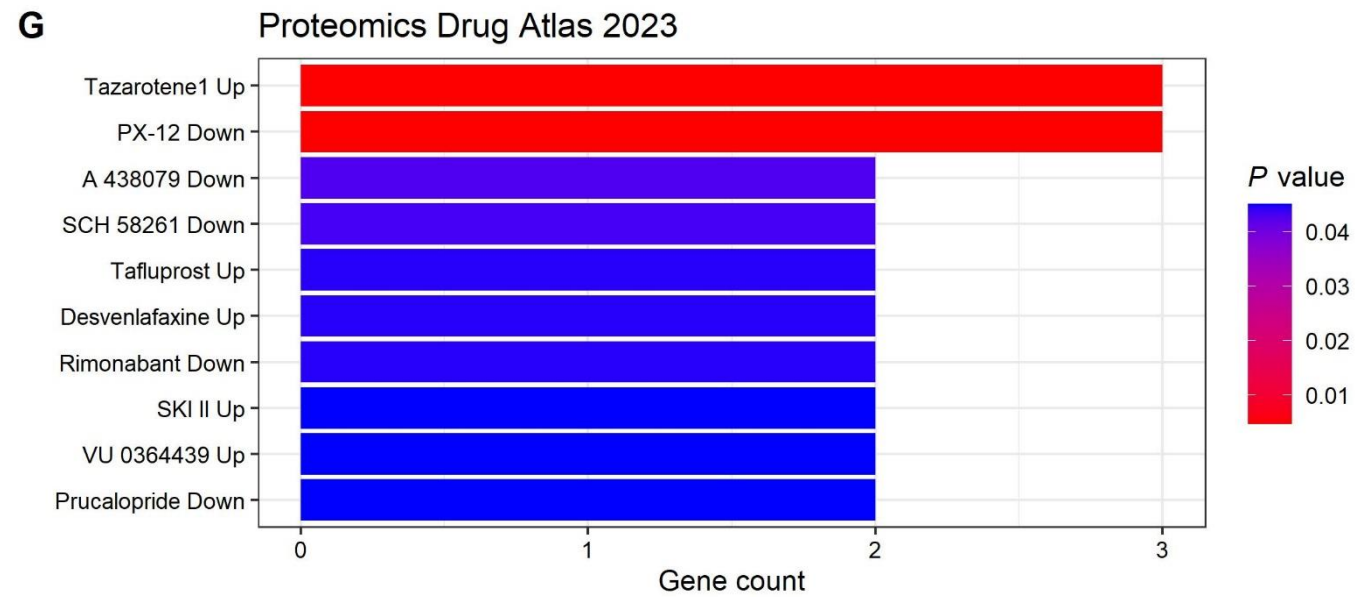


Figure 3. (G) DrugAtlas drug-gene interaction results.

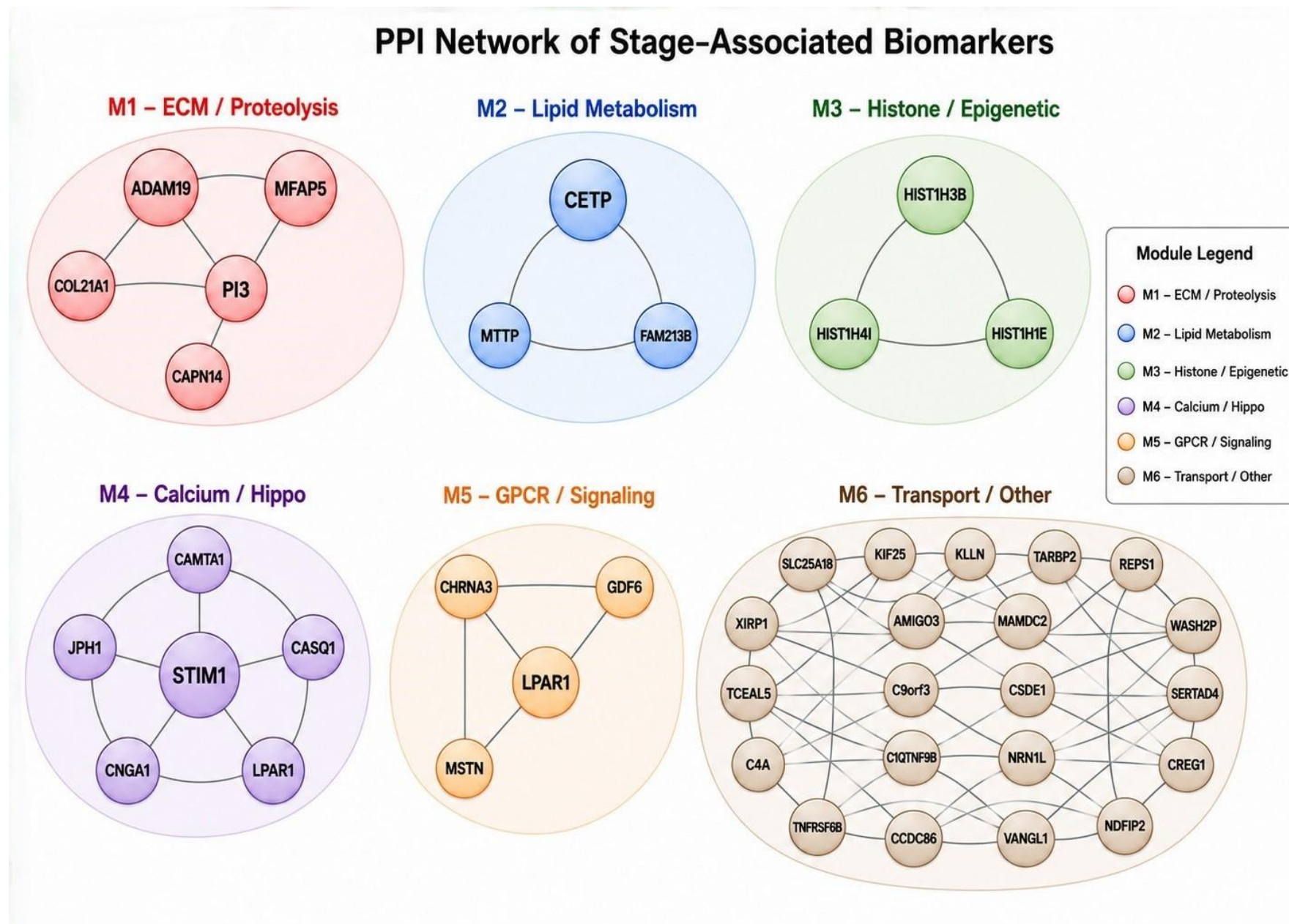


Figure 4. PPI network and module-based functional analysis of stage-associated candidate biomarkers in skin cancer progression.